COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 1 : Session – 11 : SLO – 1 | EX-OR gate
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I. Match the Following

• 1. EX-OR Logic Expression — b) A⊕B = A■B + AB■
• 2. Output Condition — a) 0 when inputs are equal
• 3. Symbol Used — c) ⊕
• 4. Truth Table Output for (1,0) — d) 1
• 5. Major Application — e) Half Adder and Parity Checker

II. Fill in the Blanks

• 1. different 2. A■B + AB■ 3. ⊕ 4. parity 5. EX-OR

III. Activity / Describe and Perform

• Logic operation: EX-OR outputs logic 1 only when the two inputs DIFFER. Boolean expression: Y = A⊕B = A■B + AB■.
• Truth table: 00→0, 01→1, 10→1, 11→0.
• Circuit symbol: OR symbol with an extra curved line at the input side.
• Practical application: parity generator/checker in data communication and the SUM output of a Half Adder (Sum = A⊕B, Carry = A·B). Experimental verification: applying all four input combinations to the gate reproduces the truth table exactly.